RELATIVE PERFECTNESS LIFTS ALONG NILPOTENT THICKENINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A complex K' whose derived reduction modulo a nilpotent ideal is pseudo-coherent is itself pseudo-coherent. invoked at proof lines 22175-22177. Gives the pseudo-coherence half of relative perfectness.
- A complex K' whose derived reduction modulo a nilpotent ideal has finite tor dimension has finite tor dimension itself. invoked at proof lines 22175-22177. Gives the finite tor dimension half of relative perfectness.

Source reference. Original section 065Y: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14899–14899. Referenced at proof lines 22174-22175 for the identification K = K' tensor R; the section is used inside the counted span.

Source reference. Original lemma 0H76: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19498–19498. Cited at proof lines 22176-22177 for the pseudo-coherence of K'; the conclusion is used inside the counted span.

Source reference. Original lemma 0H75: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17307–17307. Cited at proof lines 22176-22177 for the finite tor dimension of K'; the conclusion is used inside the counted span.

Lemma 0.2. *Let $R' \to A'$ be a flat ring map of finite presentation. Let $R' \to R$ be a surjective ring map whose kernel is a nilpotent ideal. Set $A = A' \otimes_{R'} R$. Let $K' \in D(A')$ and set $K = K' \otimes^{\mathbf{L}}_{A'} A$ in $D(A)$. If K is R-perfect, then K' is R'-perfect.*

Proof. Observe that $A' \to A$ has nilpotent kernel and that by flatness of $R' \to A'$ we have $K = K' \otimes^{\mathbf{L}}_{R'} R$ (see Section 065Y). Hence the lemma follows by combining Lemmas 0H76 and 0H75. □

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